

Project Design Phase

Proposed Solution Template

Field	Details
Date	18 June 2026
Team ID	24EG105H22
Project Name	HouseHunt – House Rent Management System
Maximum Marks	2 Marks

Proposed Solution

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Tenants face difficulty finding verified, affordable rental properties while property owners struggle to reach genuine tenants and manage inquiries effectively. Traditional methods (brokers, social media, classifieds) are fragmented, slow, unreliable, and lack transparency in the rental process.
2	Idea / Solution Description	HouseHunt is a full-stack House Rent Management System that connects property owners, renters, and administrators on a single secure platform. Owners list properties; admin approves them; renters search, filter, and submit booking requests; owners accept or reject bookings. Each role has a dedicated dashboard. Built with React.js + Vite (frontend), Express.js (backend), PostgreSQL + Drizzle ORM (database), and JWT authentication.
3	Novelty / Uniqueness	1. Three-tier role system (Owner / Renter / Admin) with distinct dashboards. 2. Admin approval workflow ensures only verified properties appear publicly. 3. Structured booking request lifecycle: Pending → Approved/Rejected. 4. Contract-first API design using OpenAPI spec + auto-generated Zod schemas and React Query hooks. 5. Zero broker dependency – direct owner-to-renter communication. 6. Built with

S.No.	Parameter	Description
		modern PERN stack with type-safe, codegen-driven API layer.
4	Social Impact / Customer Satisfaction	• Reduces time and cost for tenants searching for rental homes. • Eliminates broker commissions saving tenants 1-2 months rent. • Increases property owner reach without advertising costs. • Admin verification builds a trusted, safe marketplace. • Improves housing accessibility in urban areas by making verified listings freely available. • Empowers both parties with transparency in booking status.
5	Business Model (Revenue Model)	• Freemium Model: Basic listing is free for owners; premium listing (featured placement, photo gallery) is paid. • Subscription: Monthly/annual plans for owners with multiple properties. • Transaction Fee: Small percentage commission on successful bookings. • Admin Dashboard Analytics: Data-driven insights for real-estate agencies as a paid service. • Future: Tenant background verification as a paid add-on.
6	Scalability of the Solution	• Stateless JWT authentication – horizontally scalable without session servers. • PostgreSQL with Drizzle ORM – can be sharded or moved to managed cloud DB (AWS RDS, Supabase). • React + Vite frontend can be served via CDN for global scale. • Express API is containerizable (Docker/Kubernetes) for microservices evolution. • OpenAPI contract-first design allows new client apps (mobile, desktop) without API changes. • pnpm monorepo structure supports adding new services (notification service, payment service) independently.